

ActInf Textbook Group ~ Cohort 1 ~ Meeting 14

(Welcome back, starting of Chapters 6-10)

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okay we begin welcome everyone it is the resumption of our cohort update the video description it's the resumption of cohort one of the active inference textbook group it's our 14 okay could someone mute Giuseppe or feel free to say whatever you want to say well uh can you hear me yes yes okay so sorry but I'm completely new to this interface of a gather and uh encoder so I I'm not sure I understand cohort one and cohort 2 here is that so because cohort one was the wasn't it the previous one like the the one that yes the start last year well it's still going on from May till July we did chapters one through five okay now we are in this cohort so if those who did not participate in this part like respectfully you know thanks for the question but everyone in general respectfully let's try to have people who are in the cohort participate in it everyone's welcome to join and of course re-watch but contributions in the questions and the chat are probably best so that we can respect the people who are continuing on this is our 14th meeting and now we're heading into the second five chapters of the book okay does any other person want to give oh yeah go ahead hold on so the the the the current one that is going on right now is the cohort one and not cohort two we're all learning in different stages so the previous hour was the cohort that is focusing on chapters one through five okay stage they're in and now some of us a different set of people some are joining both so that's why we made the meetings convenient was so that we could like for many of us who want to do it both we can join or there's but this is a separate set of people I see I see yes so okay so cohort true was was first and then I mean right in the this meeting in this meeting right now cohort 2 was first yes and now which started correct before we warmed up the chapters one through five okay so I read it because I I subscribed for cohort too so excellent okay good thank you so much see you next time thanks for setting up all this yeah this thing okay bye bye yes awesome okay anyone else want to add anything we don't need to do a coda overview I would say people who are watching this recording can look at the live recording if they want a code to update that we just went through for the cohort twos live meetings but does anyone

want to give any um introductory thoughts otherwise there's a few ways I think we can go for this session but we're here just to rejoin our activities and Orient towards chapters six or ten we can scan through the chapters we can have any suggestions or discussions around ways to approach them I don't know uh Rohan and then anyone else I was wondering when are we gonna get to the projects part or discussions about the projects great question let's look at the chapters and see where that first off it can happen however you want with contacting people and um like we can make a sub page like I see exists here this looks really awesome so people can request a sub page they can mention it that contact them now or however so it's up to you for a project that is under your control it can be developed in the open here or however you can also participate in many ongoing projects like active block fronts which also um which might all share this are get coin Grant was approved yesterday so if people are familiar with this or want to become familiar this is a way to like provide or receive support possibly for their learning in blockburns and contributing to that open source package [Music] um does that address what you were thinking were you thinking about these project ideas or were you thinking more about like the sort of build your own model approach that will be starting in chapter six I know about uh project ideas in the quota page awesome yeah like these are well they're they're different different types of projects clearly more will be added but they're of different they're not necessary they might involve building a generative model but they are not degenerative model itself or only let's look at chapter six because this is where the generative model construction is going to come into play and so again there's a lot of projects that you would only need to qualitatively think about generative models for like educational material production is related to the generative model or could be modeled as a generative model of a learner but you wouldn't necessarily have to make a general model but in chapter six it's a recipe for Designing active inference models it is going to rock sorry just real quick uh I think Luke is here uh doesn't have access against or access okay okay message in Discord or I mean messaging gather your email address and I'll add you Okay so sorry I didn't oh okay you requested access all right I'll add you that's enough then in chapter six it's going to ask some questions that are uh driving the design of the generative model which system of remodeling system of Interest what is the appropriate form for the reflecting the discrete or continuous distinction that we looked at and also all these other questions that aren't exactly within the textbook like temporal depth hierarchical nesting which parameters are learned versus fixed all these different directions that the models actually have to be elaborated on even before getting to the data engineering and maybe even code implementation just from a structural perspective

what form is it how to set up the generative model is where some of those analytical details begin being introduced not necessarily even the program level implementation or an executable simulation but like okay we have a preference Vector that's is that is there two states are is there some continuous preference distribution whatever it is what is the actual details of the and then the other side of the coin slash blanket is the generative process so what are the endogenous Dynamics and the um causal mapping with action for the generative process and there's kind of cases where you have more of a communication setting active inference entities on both sides you could also have more of like a simple simple partnering like two um deterministic programs that send messages to each other or it could be like one more cognitive more agent-like generative model and one more environmental non-living type any thoughts or overviews or how would people think about how can we make this type of a recipe design should we be all designing a model or how do we even go about reading a section of the book in a chapter that begins with a recipe provision Lyle yeah so so I I spent a lot of time building what I think would be classed here as generative models and what I find is that this what I I characterize it as the step of framing it up framing up is is often far trickier sometimes much harder frankly than building a model depending on what what the situation is and in my experience I find that it all comes down to making sure we understand what is the question that we're trying to answer right it's sort of like because because every modeling process I'm sure everybody on this call knows is the process of deciding what to leave out right what to put in what to leave out and what level of representation then it all has everything that's in there if it's going to be a good model has to be necessary and necessary to answer a question or provide a particular Insight or so forth so that whole issue of framing out uh what you're going to build you know and in all those different the 10 000 questions yeah there's more than four you know ultimately it's many then it's all about that which often means understanding the audience if if the model builder is the only audience this is a much simpler if the audience is uh you know some much larger group with different perspectives a lack of of homogeneity and insight then there almost has to be uh I'm gonna say almost a pre-modeling or framing process of of sort of of framing up what are the questions we care about and then we can talk about how how those translate into a model so that's just some general framing thank you that's very interesting it it um makes me think about like this audience question for the model like why are we building the model why are we shaping our regime of attention to build this model building can be an end of itself it can be creative and expressive and fun and all these other things and maybe later there's a road that has some pragmatic value too but also

it can just be like fun like you know what would a generative model that could do this look like and that can be very didactic there's other times where you're presenting information oh I made a simulation of Twitter discourse and dot dot dot is always going to be presented in a way that's not only just like the model distilling down from the world but then you're communicating it and then when you're actually not just communicating but if you're making an application that does something and communication is doing something as well then the system of Interest as Lyle mentioned is totally non-trivial and there's the whole framing in some of these early stages especially because they can't always be changed easily later um run and then anyone else uh yeah uh so I'm just curious when you say who's the audience uh is it that uh like isn't this part of the active inference model itself that okay so whatever makes sense in that framework wouldn't that be the audience here thank you as in it has to survive right your agent has to survive so whatever generic model helps it survive would that not be the model of choice over here and I'm just curious I'm not sure that's awesome it's awesome like way to frame it and it's it's um maybe someone can share the um free energy governance link in the in the chat I just can't look it up right now but but um it puts this very nicely in in a recent pretty short book dissertation about like that imperative for organizations and um yes in one sense it can be said the right generative model or like the audience for it is whatever its survivability conditions are but and within our textbook group The survivability conditions are open however we're also learning about or even just gesturing towards like not just we're gonna imagine a room with a person who can take on a jacket or take it off um we can make that a didactic model and the audience could be this textbook group or future cohorts of the textbook group and the function could be just sort of like a neutral Exemplar active inference generative model but then of course the considerations for information presentation empirical data analysis any kind of application the system of interest is not just the one person with the jacket on or off two states room with a sign fluctuation so that's a different situation for model building so is that how should we interpret this recipe oh okay okay but is there not like a journal framework for what a good generative model looks like is what I'm saying like I I guess you could frame that definitely so if you want to build pixel level representations for example you can use again and that's a pretty decent one but maybe pixel level representations are not a good uh not good for the task at hand or something like that so it should be able we should it should fit a general category right so this is what a good set of models looks like for active inference so that that sort of is getting it yeah that's a very fascinating framing does anyone want to add something to that is I I totally hear you like on what is there a is there a

rubric like what are we evaluating the generative model and process fit because it's not like there's just simply a best generative model you know bigger or you know whatever no future about it except outside of a generative process um context is the fitness and then what is their rubric is our gold standard is it like oh you're dealing with audio data here's the file format that level of standardization certainly however are there some um practices or lenses that we could bring in from people's experience with evaluating model adequacy and relevancy there probably are Lyle well yeah that really comes right back to purpose right so a model is always fit for purpose and and you of course this also surfaces the questions of model validation right one of the things we haven't talked about here is what's the right model how do we frame up the right model what do we choose to put in what leave out part of that is purpose and and how would I evaluate you know if I do what I'm imagining a model will be how do I evaluate its quality its validity well what data is actually available that's one thing you always have to ask at the outset what data is available what data can I use what's computationally you know adequate and what parts of this then uh when we kind of project you know when we apply this model if that's its role then does it fit for the purpose does it provide you know good you know adequate results and improve to be you know whatever Behavior we're trying to address you can't you can't say if it's good if you don't have a purpose so if your purpose is just to explore model building which is something I love to do then that's a pretty big there's a lot of things that are fit for that purpose right but if the purpose is to um uh you know Pro do a prospective uh analysis of a system and what actions might be taken to intervene then that's in a different thing thanks a lot for sharing it I'm I'm really happy and I think it's relevant that we're having this kind of a discussion because we've gone through some of the first chapters and we will continue through chapter six I think into having this conversation but does anybody else want to add something about this more General topic of like how are we even coming to talk about what system or how are we framing the system that we're modeling that's in section 6.3 I um I mean this is too broad to it's not active in French specific but whenever I'm trying to model something the only questions that I mean it is contextual to the purpose it entails a purpose in an audience but all I want to know basically is what are the questions that need this model needs to like generate an answer for um what is the information that is like necessarily therefore kind of required to Model A system that would have such answers in it uh and maybe another way to say the same thing is that like what affordances would the model acquire to answer address whatever the some set of questions that are posed to the system so if you asked you know I don't know um like how frequently does a yellow colored car I don't know drive

down this particular Road um then I wouldn't need to record like the number of bikes or their colors uh that are you know using the bike lane or buses or trucks large you know so and like this is um maybe outside the scope of what I mean by yellow how many you know cars out of all the cars how many are yellow or something that's right driving down this road or whatever um it's like not a even exactly a generative model but my point is that modeling that system there there is obvious data that's like completely irrelevant right um and so if you were going to build some sort of simulation of any kind that's going to be generative like there's often a lot of data that's just completely extraneous depending on what questions you're going to ask of it thanks in six three in the first paragraphs they're gonna come out hard on the interface concept of Markov blanket this is how you'll do system modeling is by finding interfaces that distinguish the internal and external states of a system is critiqued or elaborated because there's an infinite degrees of freedom in the core screenings and those are modeler specific and they suggest that that matters especially if you want to be considering embodied extended perspectives where just descriptive interfacing models won't give you the whole answer there's a great paper could a neuroscientist understand a microprocessor showing that even if you have total access to the microprocessors firing like an EEG and a generated set of single and double lesions a lot of the interpretations would be strongly misleading um yet people want the wiring Circuit of the brain which is dynamic and also involves diffusive molecules we have all 302 nematode neurons mapped but that also as an interface diagram is not even close so if you want to describe the nematode like model building then it's a fit for purpose however there might be very simple models that could be doing a lot better in this in nematode Behavior observation the form is discussed there are three main design choices first discrete and continuous time we had some awesome discussions in the previous months about like similarities and differences and where like in theory they're different and then in practice where they're different the whole question of like discretizing continuous systems um the second choice is between shallow and hierarchical models where inference operates at multiple time scales the third choice is the consideration only of the present kind of the variational free energy versus models with temporal depth temporal depth consideration could be purely passive and it could also include some elements of affordance which if we remember to like the B Matrix actions are influencing how States the world are changing through time so that is how active inference is unifying policy comparison by kind of embedding action into estimates of how things could evolve probabilistically those three design decisions are explored in slightly more detail in the subsections six five how to set up the generative model

after specification then the specific variables have to be included okay the latent State temperature is going to be continuous the observations in my thermometer I'm going to take in as just integers and then there's going to be a person who is going to be in a binary state of like whether it's hot or cold or maybe I want hot neutral cold or maybe I want them to have a 10 point scale or I want them to have um a scale that's doing Dot that is in this stage of actually setting up the generative model and giving a little bit of like color to what had been just outlined it's going to be a nested model with this continuous variable and this kind of a discrete variable here we return to the Frog example from earlier in the book where there's like but it's kind of being generalized a little bit it was introduced with the whole Bayesian equation walkthrough now we're returning to it and thinking more about multimodal or like even you could consider this as Sensor Fusion because there's some hidden cause the jumping or the action of a frog that is inducing several domains of impact potentially in the generative process and then that leads to specific observables from the perspective of the cognitive entity those could be again two communicative cognitive entities or one and not or so this is like okay we're going to be modeling forest fires so forest fires they change the temperature they change the chemical composition they change the vibrations in the ground or something okay and then maybe there's three sensors maybe you have three thermometers and then what about them are you gonna average them are they going to be like three parallel modeled Time series that is where some more specification of the generative model comes into play keeping this the active inference ontology in mind what are the affordances the Frog what are the latent States that are being changing through time like sound physical presence or Photon emission or something and then what are the data that would be oh the observations this box looks like it's about different sensory modalities I'm sure there's a lot of interesting discussions we could bring in here priors and empirical Behavior complete class theorems any statistical decision procedure Behavior may be framed as Bayes optimal under the right set of Prior beliefs discussion of fixed and learned variables one can imagine the pros and the cons generally of having fixed and learned computationally it's simpler to fix a hard-coded variable however if there's a setting where the model's adequacy depends on being able to update that maybe you want it learnable however then you have to deal with these questions about like the kinetics of learning like instantly switch or slow switch or Trend averaging so there's a lot of discussions around learning how is attention regulated what is precision on learning it's like it very much complexifies the and there's many variables that you can bring into a learnable setting but when it's framed as just static and only like the hidden States changing through time that's like

the simplest case but more and more cognitive entities probably are going to entail considering learning Dynamics setting up the generative process the reason we postponed the design of the generative process a little while of hands entity first not the um Niche first is that in many practical applications discussed in this book we simply assume the Dynamics of the generative process are the same or very similar to the generative model we generally assume the agent's generative model closely mimics the process that generates its observation that is clearly a simplifying case because it doesn't address what if you're paying attention to predicting with and the world is dependent on height there's so many complexities of that which is really that broader like what is being modeled what matters question but in some way very upstream or very close to the Stream it is like the rabbit being put in the Hat like if the rat knows that there's a 5x5 maze that's a huge amount of information versus if it only has access to Pure local information and uh that that's just something to I think for us to note as we look through models like are we modeling the generative process oh so the seasons oscillate so it is a sine wave so then the generative model of the agent is also going to be fitting a sine wave well then it kind of collapses to like a stochastic regression and maybe then it looks like the fit is going to be very good active inference modeling brings in another complexity with action influencing outcomes that prevents some um I guess fallacies of modeling but by no means all it probably introduces many others Eric yeah uh can you hear me uh this is a news sounds good um so this this might be a time to to um to discuss the question I posted and I apologize I didn't put it in the right place the other day um I just figured out today that it goes under this particular location but this is about the question about why modeling of a generative process is necessary is a necessary step for building an active inference model so let me just read it um yeah because it's a lot of words uh so you know we know that an active inference agent employs a generative model as this model of the external world and it uses its General model to perform inferences about perceptual input and to make decisions about actions so in general the external world is complex and unknowable to the uh to the agent the generative model may or may not be able to learn but if learning is supposed to uh if it is learning is opposed to put the generative model in a better alignment with the world's generative process whatever that may be so I can see how one might model the generative process if one is building an artificial agent in a laboratory where you kind of have access you you control how the world behaves with respect to the agent and then you might want to study how well the agent design um Works versus you know one versus another under control conditions but this is not the same as designing an agent to go out and cope with some real world environment where you really

don't know what the generative process is so I think that the recipe for designing an active interest models would include formally setting up the generative process only in the kind of a laboratory situation but not if you're really trying to build something out you know for the for the real world because you don't know the general process you might create an artificial laboratory kind of situation by trying to build a model of how the world works and say well how does my agent work under that those assumptions or not but that you know that might set you up for putting the agent in the real world but in fact you don't know the degenerative process if you're really trying to build something to to work in the real real world That's my kind of my question that's a question thank you Lyle well I I love uh Eric I love that that framing because I think there's multiple different cases right so I think if you're modeling something uh that as you say goes in in the real world then you know perhaps it's a you know some sentient creature or being um then then it could obviously there's the real world even if we knew everything about it is too complex to model right so so that's that's Fool's Gold you know to assume you know everything right there and and so I don't I'm just gonna say yeah that's hard you know typically I guess we would say what are the parts of the generative the real world degenerative uh process that we're going consider relevant and deal with that and knowing that we're making a leap of faith but there's another case I think which is is more generalizable just to come a complex adaptive systems that operate in what I call the built world all right so uh I'm I work with a bunch of people who build digital twins they build digital twins of um human infrastructure right so buildings and all this stuff they're the generative process that we're concerned with is largely knowable right and largely represent representable and so and and what's interesting about that is there are a bunch of blind spots and inadequacies in what we know about that generative process and the ability to do inference about what the real state of that system is then becomes the value of this approach in that setting right so so I think I I don't think there's a general answer once again I think it really comes down to purpose and what the specific framing you're looking to do to me this ability to say something about what I I like that idea of decision agents right decision agents that uh are going to potentially take action or infer actual States is is much richer than what a lot of the modeling historical modeling paradigms have given us so I think that's really interesting thank you Lyle Ali uh yeah and uh I also think that the distinction between uh General surprises and generative models becomes significant uh if we want to simulate learning rather than just inference uh I mean that's the answer I got from Ryan Smith because I asked him almost the same question why would why do we need generative process in order simulate the agents and

um because you see for the generative if the generative model exactly Maps them to generative process we only need that capital D Matrix right but uh and that just when we're just trying to stimulate the inference side of the agent right but when the learning comes along then we need to separate uh that capital d uh from another Matrix probably the case D which Maps into the generator process so yeah one of the situations that the distinction between two might be necessary is when we when those two process I mean learning and inference are taken as two separate phenomena to um one note is also there's work on symmetries in the generative process generative model like I think like the bi-directional paper with Matt Sims and blue um discussing it and then here will be a defense to Eric's excellent question The Entity first modeling approach that's taken by this chapter one can say I don't know whether I'm going to a forest or a desert but I'm going to have these sensors on my designed interface and I'm going to do extensive modeling on interception and homeostasis and if I can learn policies that balance homeostasis then I've survived if I fail then failure happens so that doesn't mean that it works or it's doing the right thing fit for purpose any of those aspects but one of the insights from the 4E and the embodied side is like a vast amount of our what is modelable as cognitive isn't necessarily the declarative I think therefore I am possibly even symbolic type thinking linguistic there's a lot going on and many of that a lot going on can be simplified by not even knowing about the external process maybe not even the as modeled kidney knowing about what the as modeled liver is doing so this type of a focus was like we're going to Define our entity types and the kinds of observations that they can make the kinds of affordances that they have or do empirically or that they conceptually could do that approach um is consistent with embodied biological systems and Persistence of organizations and so I'm wondering how that piece comes into play with this discussion that's very important about and also their earlier discussion of the embedded and extended cognition is that it kind of blurs the line with what is a generative model what is a generative process hence there being a lot of thinking and attention on what the system's modeling is what are the known unknowns and the unknown unknown is going to be let's just see a chapter ends six seven simulating visualizing analyzing and fitting data using active inference standard routines for active inference that provide support for all functions are available #spm and appendix C step-by-step model is a great look through this this is Matlab code one amazing work that we could all iterate on would be porting Matlab into the ontology into Python and or Julia with pi mdp or active block friends but there's a lot in SPM and some of its neuroimaging specific some of it's very variational base specific not all of it is going to be state-of-the-art or the most useful in every situation I

wonder if only offering certain routines in Matlab which is like a traditionally very academic software platform and potentially non-free can be a limiting factor so those who feel like they want to learn more or contribute this way helping structure some of the code architecture so it cohorts coming through can be doing multiple language maybe first we can just move some of these routines into a folder and then people can start to organize those types of um aspects summary it's a four-step recipe we walk through it from the consideration of the system of Interest broadly like the entity and the process focusing in on the entity specifically thinking structurally about what it is going to be like doing slash composed of specifying that to a few levels of detail then doing the same for the generative process and then once that is specified it's not quite there for most people today but where it seems like they are laying out a path towards and where we probably can and will be after that level of setup in some potentially graphical editor or parameter file as as SPM and Matlab have done in many many papers the generative model specification almost prepares the working script but there's still a lot of details and and pieces to add in especially in a modern environment and for all of these like non-standard and novel cases that people are wanting to consider but that is once we have a notebook that goes through the recipe then the recipes can be forked and so on and they can be written in different languages Pi mdp block friends Ali yeah and just one thing that I think is not made explicit in this chapter is that actually that first step which system are we modeling especially uh the identification of the marker blanket part is uh basically at this point of active inference research uh is more like a like a hypothetical step because I I also had a misapprehension that uh we can somehow uh identify the markup blankets from data but then as I went on through my project and by Consulting Maxwell and others um actually this is the comment Maxwell uh made about my proposal that he said that identifying markup blanket from data is still an outstanding issue that is not resolved yet in the uh research and active inference and one of the most active areas areas of research they're doing at versus lab is just focused on this part of the modeling recipe I mean how to how we can identify automatically identify or somehow learn the markup blanket from just pure data points awesome and that ties in with Dalton s's work on the blanket index and like continuous variables for dynamic Assemblies of blanketing like it this is gonna get really nuanced and empirical with the discovery and the thresholding of Markov blankets and causal inference in Real World Systems with many different kinds of sensors and doing structure learning on generative model and process and like everything empirically which is why as a simplifying step the uh it actually really reminds me of how Dave Snowden situates the difference between systems

thinking and complexity thinking which is systems often uses like a lot of nodes and flow charts to represent system elements and relationships and like interfaces whereas in complexity in the canevas especially um framing it's like more like self-organizing sand dunes and so causal relationships might be very much in flux and it puts a higher focus on action rather than system description especially in situations like moving your head where the relationship of different things is going to be heavily action dependent um and so for now we're kind of in the systems world where it's like a priori here the system interfaces and then we're going to take whether it's a heuristic or whether you take a literal interpretation but you don't have to you could just say I'm taking heuristically the approach of the systems mapping but that's not to say that that generative model couldn't be trained in a systems mapped context and then placed into a different setting where there was a mismatch or a different generative model and so it's going to be an ongoing discussion how are when are these lines drawn one other point um other than not needing to use last names to refer to specific Concepts um like are we talking about the blanket that's statistically insulates every partition of nodes in these base graphs which are going to be vast as Ali and Jacob and I were discussing on the um branching time active inference from uh just a few days ago like these graphs are going to be vast and so just saying there will be a markout blanket in fact there will be many Markov blankets with different semantics that you might assign to them or not how does that map onto the real world is the realism instrumentalism pragmatism abduction conversation that we've been having do you come in with these kinds of boundaries heuristically and distinctions or how many levels of uncertainty in nesting I don't know how many categories of blank there are I don't know if it's a category or a Continuum of this you can actually still take on multiple levels of uncertainty in the modeling but then fitting it with empirical data if it's fit for purpose use is to do some sort of statistical inference or like provide meaning or action but just didactically when the model's purpose is educational then there's a lot of things in architectures we might play with and then the recipe and also this conversation uh helps I hope hold space for the educational and the Curiosity driven model development where we're going to have a lot of fun and interesting GMS and people's applications and working with different real world constraints in different settings we'll all be learning a lot from it and hopefully some of it will be in repos we can reference other people may want to play around inversion within Boulders that we can make if anybody wants to be added as a collaborator in this textbook repo which is just pretty much empty except for I guess a script from Jacob and there'll be more to say on probably GitHub and on um as we continue through but in these last minutes

does anyone have any thoughts for our continuation of the cohort I have one one question uh Daniel you referenced a paper earlier in the discussion that I'd love to get the reference to about uh Persistence of organizations and survivability oh yes um all right I will put it into into resources here here's the work it's a PhD dissertation that was published as a book and I'll upload the PDF as well this is uh extremely work and okay yeah just it just came out and in our last minutes it's too too fun not to share foreign has written the forward what is a firm's purpose what is an organization's purpose it's as difficult to answer as why am I here free energy governance and the fep inverts the question instead of asking how a system how must the system behave to survive it asks how do systems that survive behave the answer is rather deflationary they survive their purpose is just sustainability some classic fristianisms um some very nice framings on page 95 there's uh some great juxtapositions so it it's always awesome to see this kind of quality who knows what people here will see and I don't see any scripts here that's the textbook but who knows maybe we can do some generative modeling of organizations or we can specify it and iterate on it realize it in Block friends and pi mdp and Forney lab and jf's symbolic sweet so there'll be a lot of fun things to explore this cohort any other questions or ideas that people want to share thanks and congratulations to people for joining this continuation it's a great continuance well I know that this part is going to be very interesting the next three months will be